AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

CATION BED

WATER SOFTENING / DEMINERALISATION SYSTEM

REMOVAL OF CATIONS FOR HIGH PURITY WATER

FRP / MSEP / MSRL VESSELS | 100 LPH – 10,000 LPH

- High quality treated water
- Strong Acid Cation Resin (SAC)
- FRP, MSEP & MSRL options
- Automatic / Manual operation
- Compact and robust design
- Easy regeneration & maintenance
- Suitable for industrial applications



FRP CATION BED
(100 LPH – 5,000 LPH)
With Multiport Valve (MPV)



MSEP CATION BED
(1,000 LPH – 10,000 LPH)
MS with Epoxy Lining



MSRL CATION BED
(500 LPH – 10,000 LPH)
MS with FRP Lining / Rubber Lining

PROCESS FLOW DIAGRAM – CATION BED SYSTEM



APPLICATIONS / INDUSTRIES



WORKING PRINCIPLE

Cation bed contains Strong Acid Cation Resin (SAC) in H^+ form which exchanges hydrogen ions (H^+) with positively charged cations such as Calcium (Ca^{2+}), Magnesium (Mg^{2+}), Sodium (Na^+), Iron (Fe^{2+}/Fe^{3+}) etc. present in raw water. The cations are retained on the resin and equivalent H^+ ions are released, resulting in softened or demineralised water (when used in combination with anion bed or mixed bed).



KEY FEATURES

- High exchange capacity and long resin life
- Removes hardness (Ca & Mg) and other cations
- Available in FRP, MSEP and MSRL construction
- Flow range: 100 LPH to 10,000 LPH
- Low pressure drop and high operating capacity
- Automatic / manual operation options
- Regeneration with HCl or Sulphuric Acid
- Compact skid mounted design
- Suitable for continuous operation
- Custom design as per water quality requirement

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet (Raw Water)	Typical Outlet (After Cation Bed)
pH	-	6.5 – 8.5	6.0 – 7.0
Total Hardness (as $CaCO_3$)	mg/L	100 – 1,000	< 5 – 10
Calcium (Ca^{2+})	mg/L	20 – 300	< 1 – 2
Magnesium (Mg^{2+})	mg/L	10 – 200	< 1 – 2
Sodium (Na^+)	mg/L	10 – 200	< 5 – 10
Iron (Fe)	mg/L	< 1	< 0.1
Manganese (Mn)	mg/L	< 0.5	< 0.1
Turbidity	NTU	< 5	< 1

OPERATING PARAMETERS

Parameter	Value
Service Flow Rate	As per table
Operating Pressure	2.5 – 4.0 kg/cm²
Maximum Pressure	6.0 kg/cm²
Operating Temperature	5 – 45 °C
Bed Depth	800 – 1,200 mm
Free Board	300 – 500 mm
Regeneration Chemical	HCl (5–10%) or H_2SO_4 (3–5%)
Regeneration Time	60 – 90 minutes
Slow Rinse	30 minutes
Fast Rinse	30 minutes
Operating Bed Volume (OBR)	10 – 40 m³/m² (depending on water quality)

TECHNICAL SPECIFICATIONS – CATION BED (100 LPH TO 10,000 LPH)

Model	Flow Rate (LPH)	Vessel Type	Vessel Dia (mm)	Shell Height (mm)	Resin Quantity (Ltr)	Service Flow Rate (m³/hr)	Backwash Flow Rate (m³/hr)	Inlet / Outlet Size (NB)	Operating Bed Volume (m³/m³)
AQU-CB-0.1	100	FRP	200	1200	25	0.1	0.3	15	10 – 25
AQU-CB-0.25	250	FRP	300	1400	50	0.25	0.6	20	10 – 25
AQU-CB-0.5	500	FRP	400	1650	75	0.5	1.2	25	10 – 30
AQU-CB-1.0	1,000	FRP / MSEP	500	1800	125	1.0	2.0	32	15 – 30
AQU-CB-2.0	2,000	FRP / MSEP	700	2000	250	2.0	4.0	40	15 – 35
AQU-CB-3.0	3,000	MSEP / MSRL	800	2200	400	3.0	6.0	50	15 – 35
AQU-CB-5.0	5,000	MSEP / MSRL	1000	2400	600	5.0	10.0	65	20 – 40
AQU-CB-7.5	7,500	MSEP / MSRL	1200	2600	900	7.5	15.0	80	20 – 40
AQU-CB-10.0	10,000	MSEP / MSRL	1400	2800	1200	10.0	20.0	100	20 – 40

Note: Above specifications are indicative and can be customized as per water quality and site conditions.

TYPICAL BILL OF MATERIAL (BOM)

Sr. No.	Description	Quantity	MOC
1	Cation Bed Vessel (FRP / MSEP / MSRL)	1 No.	FRP/MS (Epoxy/Rubber/MS FRP Lined)
2	Strong Acid Cation Resin (SAC)	As per capacity	SAC (H^+ form)
3	Multiport Valve (MPV) / Butterfly Valves (MSEP/MSRL)	1 Set	UPVC/CPVC/PP/CI/SS-304/SS-316
4	Top & Bottom Distribution System	1 Set	UPVC/CPVC/PP
5	Piping (UPVC / CPVC / MS / SS – as required)	1 Set	UPVC/CPVC/PP/CI/SS-304/SS-316
6	Pressure Gauge	2 Nos.	SS-304 / Glycerine Filled
7	Sample Valve	1 No.	UPVC / SS
8	Air Release Valve (ARV)	1 No.	UPVC / SS
9	Drain Valve	1 No.	UPVC / SS
10	Flow Meter (Optional)	1 No.	SS-304 / UPVC
11	Control Panel (Auto / Manual) – Optional	1 No.	MS with Powder Coating
12	Chemical Dosing System for HCl / H_2SO_4 (Optional)	1 Set	PP / PVDF
13	Regeneration Tank (Optional)	1 No.	FRP / MS (PP Lined)
14	Interconnecting Fittings, Supports & Accessories	1 Set	UPVC / CPVC / MS / SS

MATERIAL OF CONSTRUCTION (MOC)

Item	MOC Options
Vessel	FRP / MS (Epoxy Lined) / MS (FRP Lined)
Piping	UPVC / CPVC / PP / MS / SS-304 / SS-316
Valves	UPVC / CPVC / PP / CI / SS-304 / SS-316
Distribution System	UPVC / CPVC / PP
Multiport Valve (FRP)	UPVC / ABS
Instrumentation	SS-304 / SS-316
Control Panel	MS with Powder Coating / SS-304 (Optional)
Resin	Strong Acid Cation Resin (H^+ form)
Gaskets / O-rings	EPDM / Viton
Supports / Skid	MS with Powder Coating / SS-304

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WTP | STP | ETP | ZLD | FABRICATION | SPARES | AMC | RETROFIT | AUTOMATION

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CLEANER WATER GREENER TOMORROW

1. FRP CATION BED SYSTEM – 1,000 LPH

SITE IN PROGRESS (FRP VESSEL WITH MPV)



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Commercial Building Project – 1,000 LPH

FRP Cation Bed with Multiport Valve (MPV) – Installation in Progress

2. MSEP CATION BED SYSTEM – 5,000 LPH

SITE IN PROGRESS (MS VESSEL WITH EPOXY LINING)



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Industrial Project – 5,000 LPH

MSEP Cation Bed with Epoxy Lining – Ongoing Installation

3. MSRL CATION BED SYSTEM – 3,000 LPH

SITE IN PROGRESS (MS RUBBER LINED)



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Power Plant Project – 3,000 LPH

MSRL Cation Bed with Rubber Lining – Installation in Progress

4. FRP CATION BED SYSTEM – 500 LPH

SITE IN PROGRESS (COMPACT SKID MOUNTED)



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Pharmaceutical Project – 500 LPH

FRP Cation Bed – Skid Mounted – Ongoing Installation

5. MSEP CATION BED SYSTEM – 7,500 LPH

SITE IN PROGRESS (MS VESSEL WITH EPOXY LINING)



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Textile Industry Project – 7,500 LPH

MSEP Cation Bed with Epoxy Lining – Installation in Progress

6. MSRL CATION BED SYSTEM – 10,000 LPH

SITE IN PROGRESS (MS RUBBER LINED)



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Industrial Project – 10,000 LPH

MSRL Cation Bed with Rubber Lining – Ongoing Installation